TETHYS

Two-dimensional Emitter of THz Hydrodynamic Simulation

Version 2.6.0
User Manual

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Chapter 1

Introduction

TETHYS software, now on its version 2.6.0, is hastily reaching the first point of maturity. Having passed roughly two years in the development of both the analytical models and the source code, it is now a reliable and robust software, although yet incomplete. With the prospect of a new widening of the team working for the development of this endeavour, and with the expectancy of a public release in the near future, the creation of a user's and developer's manual is of the utmost importance. However, as a research project, TETHYS source code is ever-changing and so keeping this manual up-to-date is a considerable effort that we wish that the reader may appreciate.

This small manual is intended to be a first and broad guide to the usage of TETHYS code, therefore the introduction to the physical model and numerical methods found in this chapter are but a general description aiming to get the reader acquainted with the terms used and approaches to the problem that we have followed. For a more thorough study of the physics and computational or mathematical concepts the reader will find a concise bibliography at the end of this manual and whenever any question, suggestion or commentary arise we urge the reader to send them to `pedro.cosme.e.silva@tecnico.ulisboa.pt`

1.1 Purpose and scope

The main objective of this code is to simulate and analyse a hydrodynamic description of electronic conduction on (mono-layer) graphene. At the present time limiting the simulations[1], both 1D+1 and 2D+1, to the case of gated graphene system. The physical description of such problem can be found at [1] and [2]. In the bidimensional simulation is possible to add the effects of shear viscosity and/or uniform and static magnetic field along the transverse direction, while the implementation of odd viscosity [3], also commonly dubbed as *Hall viscosity*, is ongoing.

[1] We refer to one spatial dimension simulation when the geometry and symmetry of the problem let one foreseen that the dynamics of the system will develop solely along the direction of the main electronic current i.e. along the GFET channel and that the quantities and derivatives along the transverse direction of the plane can be discarded accordingly. If, on the contrary, the fluid is expected to display a non-parallel flow or transverse variations a complete 2D simulation is required.

The equations governing this problem can be nondimensionalised which is done resorting to the channel length L, equilibrium number density n_0, velocity v_0 and the Fermi temperature $T_F = E_F/k_B = \hbar v_F\sqrt{\pi n_0}/k_B$, with $v_F = 1\times 10^6\,\mathrm{ms}^{-1}$ the Fermi velocity. Thus, writing

$$x^* \equiv x/L \quad t^* \equiv tv_0/L \quad \omega^* \equiv \omega L/v_0 \tag{1.1a}$$

$$v^* \equiv v/v_0 \quad n^* \equiv n/n_0 \quad T^* \equiv T/T_F \tag{1.1b}$$

all quantities are henceforth hence, from this point forward all the quantities are implied to be dimensionless, even though the asterisk sign will be dropped for simplicity.

The complete model that the code aims to simulate is given by the following equations, already written in the form of conservation laws;

$$\frac{\partial n}{\partial t} + \boldsymbol{\nabla}\cdot\frac{\mathbf{p}}{\sqrt{n}} = 0, \tag{1.2a}$$

$$\frac{\partial \mathbf{p}}{\partial t} + \boldsymbol{\nabla}\cdot\left(\frac{\mathbf{p}\otimes\mathbf{p}}{n^{3/2}} + \frac{v_F^2}{3}n^{3/2}\mathbb{1} + \frac{S^2}{2}n^2\mathbb{1}\right) - \\ - \nu_s\nabla^2\frac{\mathbf{p}}{n^{3/2}} - \nu_o\nabla^2\frac{\mathbf{p}^\dagger}{n^{3/2}} = -\omega_c\frac{\mathbf{p}^\dagger}{\sqrt{n}} - \frac{\mathbf{p}}{\tau} \tag{1.2b}$$

and

$$\frac{\partial T}{\partial t} + \boldsymbol{\nabla}\cdot\left[\left(\frac{T}{n^{3/2}} + \frac{3}{4\pi^2}\right)\mathbf{p}\right] - \alpha\nabla^2 T = \frac{S^2}{v_F^2}\frac{\mathbf{p}}{\sqrt{n}}\cdot\boldsymbol{\nabla}n + \Pi, \tag{1.2c}$$

for the number density n, momentum density $\mathbf{p} = m^\star n\mathbf{v}$ (with $m^\star$ the effective mass of carriers and $\mathbf{v}$ the fluid velocity) and temperature T, and where ν_s is the kinematic shear viscosity, ν_o the odd viscosity [3, 4, 5], $\mathbf{p}^\dagger = [-p_y, p_x]^\top$, ω_c the cyclotron frequency, τ the typical time for inelastic collisions, α the thermal diffusivity and Π encompasses any possible source terms for the heat flux. The parameter S can be interpreted as the sound velocity for the plasmons [2, 1] and for a GFET with a capacitance per area $C_g = \varepsilon/d$ it is given by $S^2 = e^2 d v_F\sqrt{n_0}/\varepsilon\hbar\sqrt{\pi}$, with d the distance between the gate and the graphene layer, ε the medium permittivity, and n_0 the steady state or equilibrium number density. Finally, $m^\star = \hbar\sqrt{\pi n}/v_F$ is the effective mass of the carriers, obtained resorting to a Drude model [6, 1]. The system of Eqs. (1.2) is implemented by separating the hyperbolic, parabolic and source terms in the equation and applying appropriate numerical methods for each one.

The employed numerical methods are conditionally stable, the simulation routines are robust for a wide range of the input parameters, provided that the validity of model assumptions still hold true. That is, highly degenerate electronic fluid where $T_F \gg T$; being, consequently, away from the charge neutrality point so that the carriers are only of one kind, either electrons or holes. Moreover, in the case of simulations with an applied magnetic field, it must not be high enough to induce Landau levels in the system. To ease the of compliance with such requirements, we suggest in Tab. 1.1 the range of the physical parameters of the system and their typical dimensionless values for the simulations.

Table 1.1: Physical parameters of the governing Eqs. (1.2) and typical parameters for the simulations. We assumed the average quantities to be $n_0 \sim 10^{12}\mathrm{cm}^{-2}$, $L \sim 2\mu\mathrm{m}$, $T \sim 300\mathrm{K}$, and $v_0 \sim 10^5\mathrm{ms}^{-1}$

Parameter	Designation	Typical values	Dimensionless input
S	sound velocity	$\gtrsim v_F$	$\gtrsim 10$
v_F	Fermi velocity	1×10^6 ms^{-1}	$\gtrsim 10$
ν_s	shear viscosity	$0.1 \sim 0.3$ m^{2}s^{-1}[4]	$0.1 \sim 1$
ν_o	odd viscosity	$\lesssim 0.1$ m^{2}s^{-1}[4]	$\lesssim 0.5$
ω_c	cyclotron frequency	$\lesssim 0.5$ THz [a)]	$\lesssim 10$
$1/\tau$	collision frequency	$\lesssim 1$ THz	$\ll 1$
α	thermal diffusivity	[b)]	~ 1

[a)] For $B \sim 50\mathrm{mT}$.
[b)] Assuming that the thermal conductivity follows $\kappa \sim v_F^2 \tau c_v$.

Chapter 2

Getting started

2.1 Requirements

TETHYS is written in `C++`, following the `C++11` standard. The provided automatic makefiles are designed for the use of `gcc` compiler which is, therefore, the main system requirement. For those who wish, TETHYS releases also include a `CMakeLists.txt` file for automatic compilation using `cmake` and `make` tools. Furthermore, the produced simulations are saved using the *Hierarchical Data Format* (HDF5), thus, the HDF5 `C++` API libraries are also required. The complete documentation and downloads can be found at `http://www.hdfgroup.org/solutions/hdf5`.

For the generation of graphs and plots some optional scripts of Gnuplot and Python are also provided alongside the source code.

2.2 Compilation

For the compilation of the source code a makefile can ge automatically generated with `cmake`. It is generally a sensible idea to create a empty directory to hold the executable files

```
$ mkdir build
$ cd build
```

now inside the newly created directory one only has to invoke to `cmake` the source at the parent directory

```
$ cmake ..
```

this will generate the make file and targets and so all that is necessary is to run

```
$ make all
```

this will generate two executable files: `TETHYS_2D` and `TETHYS_ELEC_2D`; for the simulation and post-processing, respectively.

2.3 Testing

Two automatic test are included by default in the cmake: the first one for the simulation itself, and the second for the electrical computations. First of all you should copy or move the test .ini file to the build directory

```
$ mv 2DTESTparameters.ini ./build
```

Then, to run them both tests use

```
$ ctest --extra-verbose
```

or, if you want to run each one seperatly `ctest --extra-verbose -I 1,1,,1` or `ctest --extra-verbose -I 2,2,,2` respectively. N.b. those test are intended to simultaneously assess all the different parameters and capabilities of the simulation code, but they do not represent any particular physical scenario.

2.4 Reconfiguring a simulation

TETHYS was designed and is meant to primarily simulate and study Dyakonov–Shur instability and therefore the boundary conditions required to excite such response are implemented. However, if the user wish to configure the boundary conditions applied to the system differently, he is expected to edit the `TETHYS_2D_Main.cpp` file according to his needs. The default boundary conditions are the following:

```
DyakonovShurBoundaryCondition::DyakonovShurBc(graph);
DirichletBoundaryCondition::YClosedNoSlip(graph);
if(graph.GetThermDiff()!=0.0){
    DirichletBoundaryCondition::Temperature(graph,0.22f, 0.22
   f, 0.22f, 0.22f);
}
```

to be applied immediately after both numerical methods. Those can be replaced by any of the provided boundary conditions[1].

2.5 Running the simulation

To run a simulation one can simply invoke

```
$ ./TETHYS_2D
```

and the program will prompt the user to provide the necessary parameters for the simulation. The necessary inputs are:

- Sound velocity S (vel_snd)

[1] See doxygen documentation for further details

- Fermi velocity v_F (vel_fer)
- Collision, of the electrons with impurities or defects, frequency $1/\tau$ (col)
- Shear viscosity ν_s (vis)
- Odd viscosity ν_o (odd)
- Cyclotron frequency ω_c(cyc)
- Thermal diffusivity α (therm)
- Saving option (save_mode)

and optionally:

- Aspect ratio of the simulation grid x:y (aspect_ratio)

where `vel_snd` is the desired sound velocity (in units of the average drift velocity) for the simulation, `vel_fer` the Fermi velocity, `col` the colision frequency, `vis` the kinematic shear viscosity and `cyc` the cyclotron frequency. Lastly, `save_mode` allows to choose between exhaustive or light run. For the value 0 only the values at the middle points of the x boundaries $n(0, W/2, t)$ $n(L, W/2, t)$ $v(0, W/2, t)$ and $v(L, W/2, t)$ are recorded. This is particularly useful for debugging purposes. Otherwise, choosing 1 the simulated data for all the spatial discretisation are recorded on the HDF5 file, saving roughly 10 snapshots per period. The user should bear in mind the me main bottleneck of the execution of this code lies, as usual, in the writing and reading of data. Alternatively, the user can pass directly such parameters as command arguments with

```
$ ./TETHYS_2D vel_snd vel_fer col vis odd cyc therm save_mode
    aspect_ratio
```

Moreover, the parameters can also be passed via a `.ini` file

```
$ ./TETHYS_2D parameters.ini
```

As standard with `.ini` files sections can be indicated by square brackets as in [section] and comment lines start with a semicolon `;` The readble keywords are indicated in Tab. 2.1

To later extract the electronic quantities from the fluid simulation data one should run

```
$ ./TETHYS_ELEC_2D hdf5_2D<...>.h5
```

2.6 Output produced

As stated before the user can opt for simplified or full output. In the simplified version the output is a data file named `preview_2D<...>.dat` organized by tab separated columns with the (normalized) values of t, $n(L)$, $v(L)$, $n(0)$ and $v(0)$ (by this order). While the

Table 2.1: Initialization file keywords

Keywords	Parameter	
sound	Sound velocity	S
fermi	Fermi velocity	v_F
shear	Shear viscosity	ν_s
odd	Odd/Hall viscosity	ν_o
col	Collision frequency	$1/\tau$
cycl	Cyclotron frequency	ω_c
therm	Thermal diffusivity	α
aspect	Aspect ratio	$Æ$
time	Simulation time	$T_{\max}$
save	Save mode	-

full output option will return a HDF5 file, `hdf5_2D<...>.h5`, with the data of density and both velocity components, besides all the relevant simulation parameters saved as attributes of the file. Each HDF5 file has a root group[2] called Data that houses the simulation attributes as well as the groups Density, VelocityX and VelocityY, inside each of these three groups the simulation results are stored for each temporal snapshot, organized in a matricial form, see Fig.2.1.

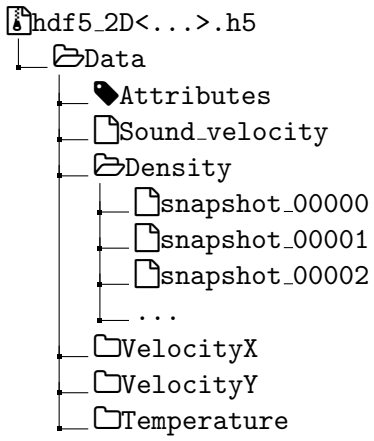

Figure 2.1: Output HDF5 file hierarchy. All the simulation attributes (e.g. time discretization, physical parameters among others) are stored directly on *Data* group. The snapshots are sequentially numbered and each one is a 2D data matrix.

The integration of the macroscopic electric quantities is done over the normalized velocity and density, and is thus also dimensionless. Here, we provide a list of the calculated quantities and respective equations and normalization constants, marked by a nought

[2]If not familiarized with the HDF5 terminology, the reader may think of *Group* as being akin to a directory

subscript.

- Net charge $Q = Q_0 \iint_\Omega n\, dxdy$, with $Q_0 = en_0L^2$
- Electric dipole $\mathfrak{p} = \mathfrak{p}_0 en_0 L^3 \iint_\Omega n(\mathbf{r} - \mathbf{R})\, dxdy$, with $\mathbf{r} = [x, y]^\top$, $\mathbf{R} = [L/2, W/2]^\top$ and $\mathfrak{p}_0 = en_0L^3$
- Electric dipole variation $\dot{\mathfrak{p}}_0 = \dfrac{v_0}{L}\mathfrak{p}_0$
- Electric dipole acceleration $\ddot{\mathfrak{p}}_0 = \dfrac{v_0^2}{L^2}\mathfrak{p}_0$
- Drain current $I_\mathrm{D} = I_0 \int j_x\, dy$ at $x = L$, with $I_0 = ev_0n_0L$
- Source current $I_\mathrm{S} = I_0 \int j_x\, dy$ at $x = 0$
- Average direct current $I_\mathrm{DS} = I_0 \iint_\Omega j_x\, dxdy$
- Average Hall current $I_\mathrm{Hall} = I_0 Æ \iint_\Omega j_y\, dxdy$
- Average channel voltage $V_\mathrm{DS} = \dfrac{v_0 L Æ}{\mu_0} \iint_\Omega v_x\, dxdy$, with μ_0 the electron mobility
- Ohm dissipated power $P_\Omega = \dfrac{I_0^2}{\sigma_0} \iint_\Omega \mathbf{j}^2\, dxdy$, with σ_0 the conductivity
- Capacitor stored power $P_C = \dfrac{v_0 Q_0^2}{LC_g} Q\dot{Q}$, with C_g the gate capacitance per area

Chapter 3

Software Description

3.1 Class diagram

The class hierarchy of TETHYS is relatively simple, it can be inspected using the class diagram patent in Fig. 3.1. The main class object of the source code main function is is always an instance of GrapheneFluid2D or GrapheneFluid1D depending ond the simulation pretended dimensionality. Those graphene classes inherit most of their member variables from the general fluid classes and override the methods of the parent class to the specifics of the graphene electronic hydrodynamics, e.g. the variable mass on the numerical fluxes. In turn, the Fluid2D and Fluid1D are inherited from the TethysBase class which is mainly responsible for managing the HDF5 file. The initialization of any of the fluid clases, be it

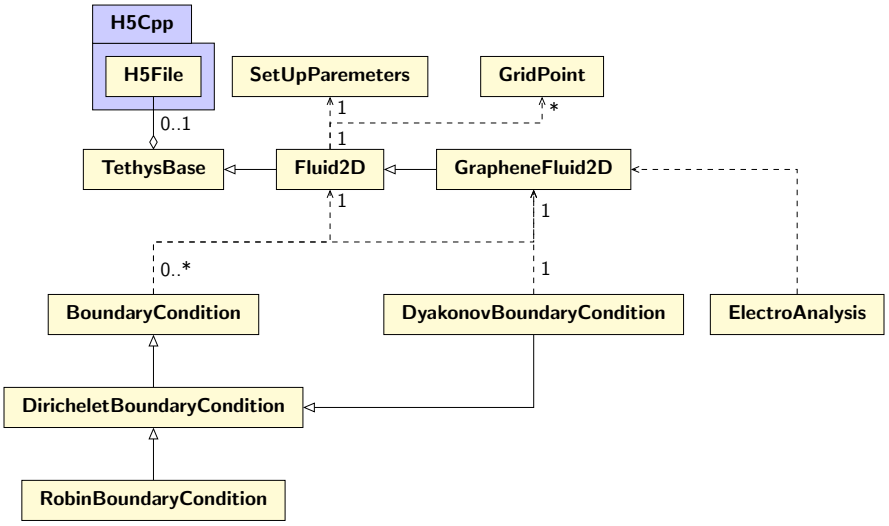

Figure 3.1: Class diagram for TETHYS source code.

1D or 2D and general or graphene, is made trough the SetUpInput class.

Lastly, the imposition of boundary conditions on the simulated variables of the fluid

classes is handled by the BoundaryCondition class and its derived classes for specific implementations. For that, a single instance of the central fluid class is passed by reference as argument to any or several of the boundary condition classes. However, the Dyakonov Shur boundary condition can only be applied to graphene fluid classes.

3.2 Execution flowcharts

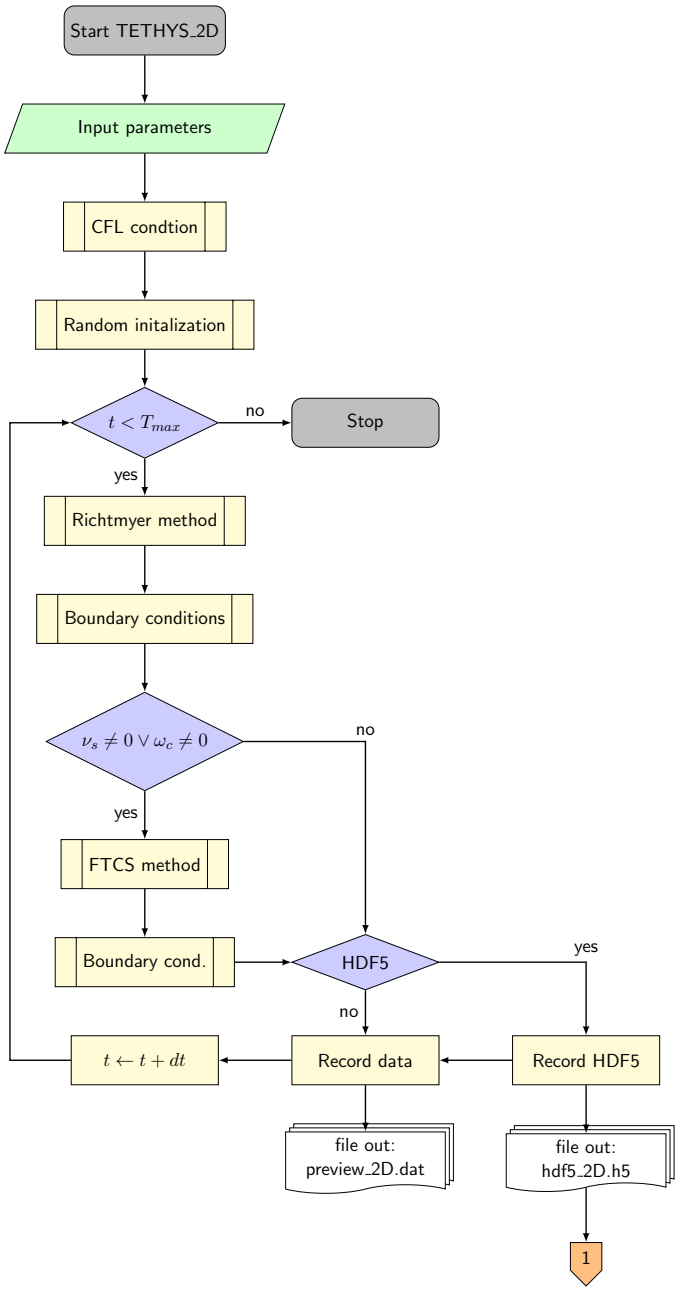

Figure 3.2: Flowchart of TETHYS_2D. Abridged description of the overall algorithm used to solve the fluid model in one dimension. The Richtmyer and FTCS methods are condensed for readability

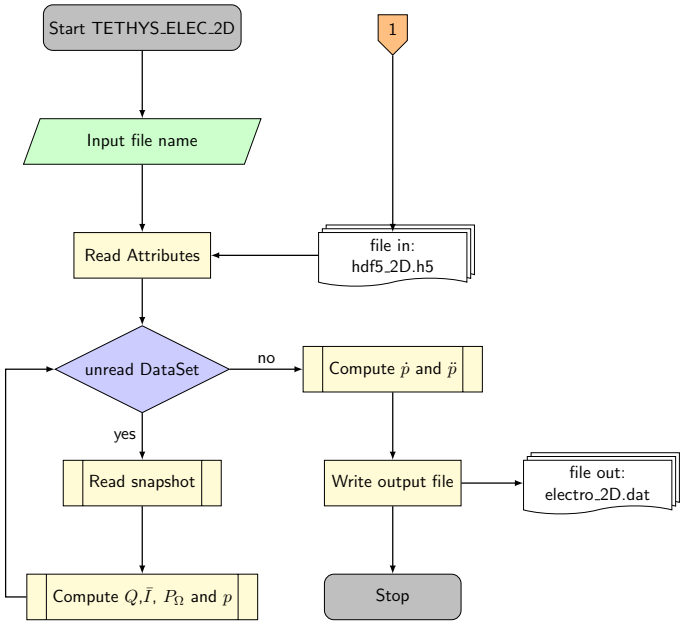

Figure 3.3: Flowchart of TETHYS_ELEC_2D. Abridged description of the overall algorithm used to solve

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